

Answer **all** questions

Question 1

Protein X is a single polypeptide consisting of 35 amino acids. It is produced in minute quantities in a particular species. The rate of transcription to synthesise the mRNA for Protein X is known to be constantly low.

A researcher has discovered that Protein X has the ability to reduce wrinkles instantly when applied topically on the skin. Realising its potential to make money, the researcher intends to produce large amounts of Protein X in a short period of time.

- (a) Suggest why Protein X is only produced in minute quantities in the particular species. [1]

1. It is under the control of a **not very active promoter / lacks enhancer**.

The genome sequence of the particular species has been determined. Thus, the sequence that codes for Protein X has been identified. The complete sequence of the non-template strand coding for the complete Protein X is shown in **Fig. 1.1** in the 5' to 3' orientation.

```

1    CTGCAGGATA TGGCTGTGCT GTACGTCAGT GGATCCGGCT GCATGCAAGA    50
51   AAGCGGCTGG ATCGATCGAA TTCTCGTGTC GTTGCTGCAC TTGCATGCTT    100
101  TGTCGAGCTC CCGTTGATGC CTGCAGCATT
  
```

Fig. 1.1

- (b) Label and indicate with two circles on **Fig. 1.1** the first and last codon of the coding sequence of Protein X. [2]

To obtain the gene of interest, the researcher contemplated between using the genomic sequence in **Fig. 1.1** and creating a cDNA library to obtain a cDNA sequence of Protein X.

- (c) Explain what is meant by a cDNA library. [3]

1. Contains DNA fragments representing only the coding region of a genome/ **genes which are expressed**;
2. non-coding regions / introns are not present;
3. involves isolation of mRNA from specific cell type / in a specific tissue at a specific stage;
4. Also, each clone has a continuous coding sequence.

Max 3

The researcher concluded that there was no difference between using the genomic sequence in **Fig. 1.1** and the cDNA sequence of Protein X to produce the functional protein.

(d) Suggest the basis of the researcher's conclusion. [1]

1. cDNA and the genomic sequence contain no introns / only exons;
2. Protein X has 35 amino acids coded for by all the 105 nucleotides (108 nucleotides including the stop codon) in the genomic sequence as shown in Fig 1.1

Max 1

The researcher decided to use the genomic DNA and the plasmid pUC18 to clone the gene.

He proceeded to isolate the genomic DNA from the particular species. Using this genomic DNA, the researcher then carried out Polymerase Chain Reaction (PCR) to obtain sufficient amount of the DNA containing the sequence shown in **Fig. 1.1**.

Fig. 1.2 shows the diagram of the plasmid pUC18 and the positions of the restriction sites found in the multiple cloning site (MCS).

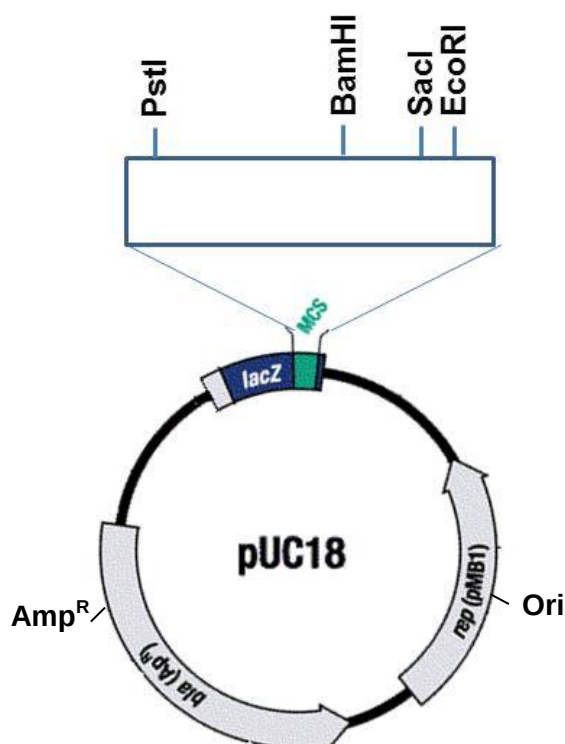


Fig. 1.2

(e) Table 1.3 shows the restriction sites of four restriction enzymes.

name	site
<i>Bam</i> H1	↓ 5'... GGATCC...3' 3'...CCTAGG...5' ↑
<i>Eco</i> RI	↓ 5'... GAATTC...3' 3'...CTTAAG...5' ↑
<i>Pst</i> I	↓ 5'... CTGCAG...3' 3'...GACGTC...5' ↑
<i>Sac</i> I	↓ 5'... GAGCTC...3' 3'...CTCGAG...5' ↑

Table 1.3

(i) With reference to Fig. 1.1, Fig 1.2 and Table 1.3, state the restriction enzyme that the researcher should use to produce the recombinant plasmid. [1]

1. *Pst*I;

(ii) Describe the steps that the researcher would carry out in the formation of the recombinant plasmid using the restriction enzyme in (i). [4]

1. Use the same restriction enzyme *Pst*I to digest genomic sequence/DNA obtained from PCR and plasmid pUC18 to generate complementary sticky ends;
2. The digested plasmid and the digested DNA from PCR are mixed together with DNA ligase;
3. the sticky ends will anneal together by forming hydrogen bonds between complementary bases;
4. DNA ligase added to seal the nicks by catalyzing the formation of phosphodiester bonds (between adjacent nucleotides on the sugar phosphate backbone), forming recombinant plasmid.

The recombinant plasmids were then inserted into bacterial cells. However, not all transformed cells with the recombinant plasmid were able to produce functional Protein X.

- (iii) Suggest why not all bacteria that have taken up the recombinant plasmid would produce a functional Protein X. [1]

1. The fragment may have annealed in the **wrong orientation**.

[Total: 13]

Question 2

Huntington's disease (HD) is a lethal neurodegenerative disorder that is inherited as an autosomal dominant condition. Due to the late onset of symptoms, patients with HD may unknowingly pass on the mutant HD allele to their children. Symptoms in individuals as well as confirmation of diagnosis in other family members are used to determine the diagnosis.

A radioactive G8 probe revealed two RFLP variants that are linked to the HD allele which can be used in the screening of HD. **Fig. 2.1** shows the two RFLP variants upstream to the HD allele and the G8 probe. The arrows show the position of *Hind*III restriction sites. The numbers represent kilobases.

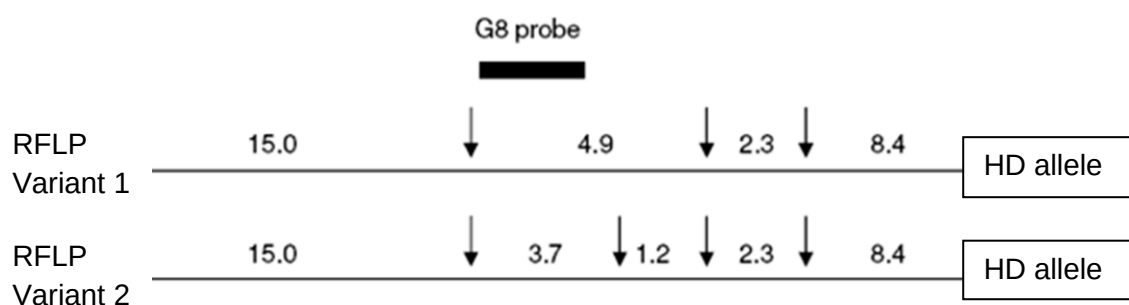


Fig. 2.1

Fig. 2.2 illustrates the Southern Blot analysis of the two RFLP variants in 30 related individuals with a family history of HD.

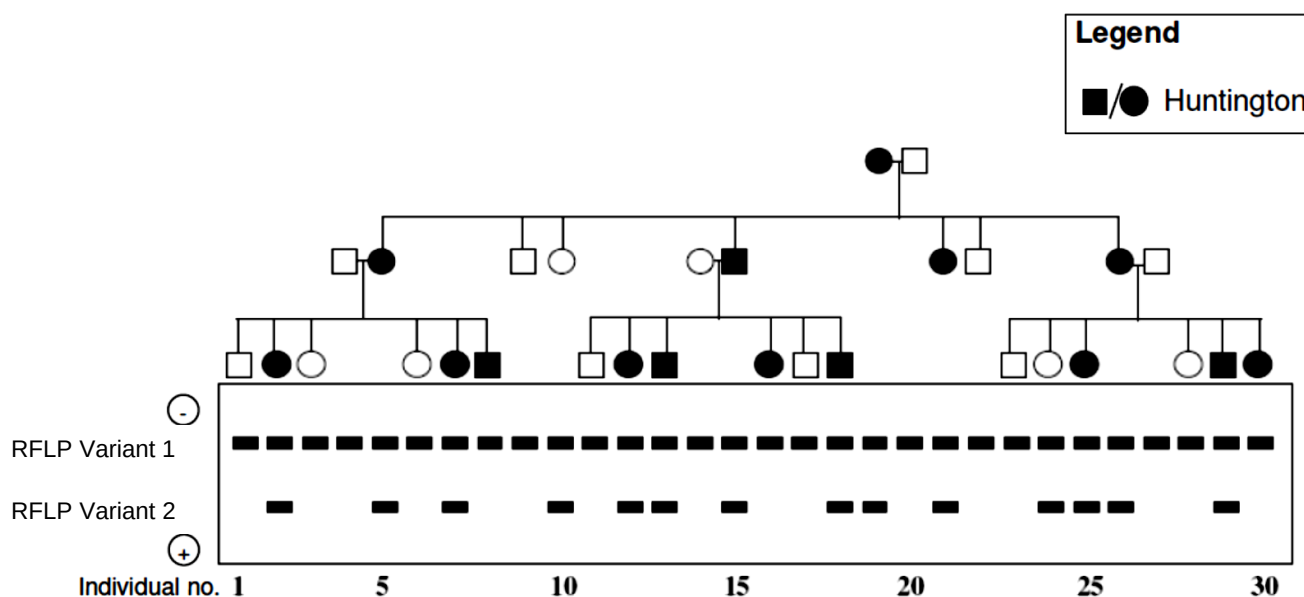


Fig. 2.2

(a) With reference to **Fig. 2.1** and **2.2**,

(i) outline the process of nucleic acid hybridization that allows the RFLP pattern of the 30 individuals to be visualized. [4]

1. DNA sample from the 30 individuals is cut/digested with *Hind*III into restriction fragments;
2. Restriction fragments are subjected to gel electrophoresis where the fragments are separated according to sizes;
3. Shorter fragments move further/faster OR cite figures (eg. 1.2kb fragments move further/faster than the 4.9kb fragments);
4. The separated DNA bands are transferred to nitrocellulose membrane by Southern Blotting;
5. Double stranded DNA denatured by NaOH / alkali;
6. Incubated with solution of G8 probe which is complementary to / which binds to / anneals to / hybridizes with / forms hydrogen bonds with 4.9kb and 3.7kb fragments;
7. Viewed by autoradiography / radioactivity exposes photographic X-ray film

Max 4

(ii) identify the RFLP variant which is linked to the mutant HD allele. Explain your answer. [2]

1. RFLP 2;

Explanation:

2. Most affected individuals have RFLP variant 2 except individual 8/16/30;
3. Most unaffected individuals do not have RFLP variant 2 except individuals 10/24;
4. Most affected individuals have a shorter 3.7kb fragment except individual 8/16/30;
5. Most unaffected individuals do not have 3.7kb fragment except individual 10/24

Max 1

(iii) suggest one possible reason for the difference in the DNA band patterns among unaffected individuals. [2]

1. During the formation of ovum/gamete in the female parent of individual 10 and 24, crossing over occurred at a region between the RFLP locus and HD locus...
2. ... cause RFLP2 to become associated with normal allele;
OR
3. Individual 10 and 24 have the mutant allele associated with RFLP 2 but symptoms have not appeared;
4. ... as disease occur later in life / as late as 40 years / person still young
OR
5. Mutation in RFLP 1 / in 4.9kb fragment in individuals 10 / 24 led to...
6. ... RFLP1 becoming RFLP2 (OWTTE) / appearance of one more *Hind*III restriction site.

Max 2

- (b) Other applications of RFLP are in the areas of genetic fingerprinting and in genomic mapping.

Explain why genes especially protein-encoding genes are seldom used as markers for genetic fingerprinting. [1]

1. sequence of protein-encoding genes are **conserved**, due to its **important** biological **function**;
2. **mutation rate lower** than non-coding region, not able to differentiate between individuals.

Max 1

- (c) Explain how PCR facilitates RFLP analysis. [2]

1. PCR is able to **amplify/ produces many copies/ produce large amount of DNA**;
2. Starting from a **very small amount of original DNA**

- (d) The human genome project holds benefits for many fields, from molecular medicine to human evolution.

Importantly, scientists are now able to test for many inherited diseases.

State **one** advantage and **one** disadvantage of the ability to test for inherited diseases. [2]

Advantages (Any 1)

1. Able to **screen embryo / diagnose disease early / before symptoms show**;
 2. **Treat person / disease, early / before symptoms appear / AW**;
 3. **Chance to carry out gene therapy**;
- Reject: ref to genetic engineering / designer babies, etc.**
4. **Allows termination / abortion / AW**;
 5. **Fewer people will have that disease**;
 6. **Informed decision making for to-be parents (e.g. about whether to have children)**;
 7. **Enables change in lifestyle / take preventive action (to delay onset of disease) ;**

Disadvantages (Any 1)

1. **genetic underclass / ref to prejudice / stigmatised by society's attitude towards genetic differences; AW**
2. **discrimination / abuse of knowledge, by employers (NOT employees) / insurance companies / banks / medical profession e.g. employer may not offer that person a job or insurance may be refused;**

Reject: ref to economic factors

3. **ref. to privacy of information**;
4. **inaccuracy of testing**;
5. **discourages people from having children**;
6. **AVP e.g. increased number of terminations / moral / psychological / social issues**

[Total: 14]

Question 3

Plant tissue culture has many applications. These include commercial production of genetically modified plants and the culturing of plant cells in a bio-reactor for the production of valuable products.

Early attempts to culture plant tissue on a culture medium were not always successful.

The culture medium contained a range of substances that were known to be requirements for plant cell growth and division. These included:

- elements, present as minerals ions
- a sugar suitable for plant cells to use

(a) Explain why a sugar was included in the plant tissue culture medium. [1]

1. **Photosynthesis not yet occurring or at low rate, therefore sugars not being synthesized;**
2. **Sugars required as energy source for respiration/ATP production/respiratory substrate;**
3. **Source of carbon for carbohydrates like starch, sucrose, and organic molecules proteins or enzymes/other suitable named organic molecules;**

Max 1

(b) Outline how plants are cloned by tissue culture. [4]

1. **Explants are surface-sterilised using dilute sodium hypochlorite to kill bacteria and fungal pathogens;**
2. **are transferred aseptically to sterile culture medium containing nutrients and plant growth regulators (auxin and cytokinin);**
3. **When equal concentration of cytokinin and auxin is added, the cells of an explant divide by mitosis to form a callus, which is an undifferentiated, actively dividing mass of cells;**
4. **Cells in callus differentiate to form root in high auxin: low cytokinin ratio and shoot in high cytokinin: low auxin ratio;**
5. **The resulting plantlets are then eventually transferred to soil.**

Max 4

(c) Explain why these plants are described as clones. [2]

1. **All cells in a population are descendants of one single parent plant, produced by mitotic divisions;**
2. **The process of mitosis involves exact replication of the parental DNA as template;**
3. **All cells are genetically identical to the same single parent, produced by asexual reproduction.**

Max 3

Investigations have shown that the original culture medium lacked key components called plant growth regulators needed for successful plant cell growth and division. Although early attempts to culture plant tissue were not always successful, some investigators have reported successful growth using root tips. Root tips are known to contain meristematic tissues.

- (d) Suggest one reason why plant tissue culture was more successful using organs such as root tips. [1]

1. Meristematic/mitotic tissue, reference to cell division activity in root tips/ rapid growth/ rapidly dividing cells;
2. Root tip region still immature tissue, thus still producing plant growth regulators e.g.auxins, cytokinins etc;
3. virus / disease, free

Max 1

- (e) Outline **one** advantage and **one** disadvantage of using tissue culture technique to produce plants. [2]

Advantages

1. allows the rapid multiplication of plants giving rise to bulk production/large numbers of genetically identical/uniform/desired plants;
2. Plants produced via tissue culture are genetically identical and all possess the desirable features of the stock plants;
3. Take up little space when compared with plants growing in fields - can be grown intensively;
4. The plantlets produced are light and small in size - can be air-freighted and transported easily/cheaply/in large quantities → increasing international trade;
5. Independent of climate changes so plants can be produced continuously/at any time of year → flexibility in meeting consumer demand;
6. possible to standardise the conditions for growth and obtain many batches of identical plants → ensures product uniformity;

Disadvantages

1. Difficulty in maintaining sterile environment/sterile conditions required. Contamination of cultures can cause very high losses in a short time.
2. High production cost, due to the sophisticated facilities, sterile laboratory conditions and special nutrient media needed / requires trained personnel with specialized skills;
3. Labour intensive → A lot of labour is required to transfer the plantlets from the laboratory to the field.
4. Problems in the transition of the plantlets → problems in the transition of the plantlets from medium to soil. Success rate of transplanted individuals is not high.

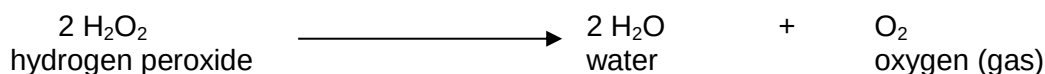
(f) Distinguish between zygotic stem cells and embryonic stem cells. [4]

	Zygotic stem cells	Embryonic stem cells
Source	<u>First 4 cells of the zygote</u>	<u>inner cell mass of blastocyst</u>
Potential	<u>Totipotent</u>	<u>Pluripotent</u>
Cell specialisation ability	Are able to differentiate into <u>any/all cell type</u> of the body, including <u>extraembryonic membranes</u> needed for development in uterus.	Are able to differentiate into <u>almost any cell type</u> of all three germ layers (endoderm, mesoderm and ectoderm).
Function	These cells give rise to form the <u>embryo</u> and then <u>whole organism</u> .	Give rise to specialized cells that form different <u>tissues/organs in developing foetus/ human body/ adult organism</u> .

[Total: 14]

Question 4

Potato catalase is found on potato tissue and will be present in extracts made from potatoes. The reaction catalysed by catalase is shown below:



The presence of inhibitors may affect enzymatic reactions. There are two main types of inhibitors, namely, competitive and non-competitive inhibitors. It is known that the presence of certain metal ions such as copper (Cu^{2+}) would result in the inhibition of the reaction catalysed by catalase.

You are required to plan, but not carry out, an investigation to determine if Cu^{2+} ions function as a competitive inhibitor or a non-competitive inhibitor of catalase.

The rate of reaction of the enzyme catalase can be measured by the rate of oxygen gas production. Oxygen can be collected by using a delivery tube from the reaction flask to a 20cm^3 gas syringe.

You must use:

- 20 cm^3 gas syringe
- 250 cm^3 conical flask with rubber bung
- Delivery tube
- Potato extract solution
- 20% hydrogen peroxide stock solution
- 5 mol dm^{-3} Copper (II) Sulphate (CuSO_4) solution
- Retort stand

You should select from the following apparatus:

- Normal laboratory glassware e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes, glass rods etc
- Timer e.g stopwatch
- Syringes
- Bunsen burner with tripod, gauze and bench mat
- Thermometer

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams if necessary,
- identify the dependent and independent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment

Mark Scheme

1. Aim: To investigate and determine if Cu^{2+} ions function as a competitive inhibitor or a non-competitive inhibitor of catalase.

2. Theoretical consideration [3 marks max]:

✓ **Theory and explanation:**

- Ref to **function of enzyme inhibitors**, e.g. enzyme inhibitors affect enzyme activity resulting in a decrease in the rate of enzyme catalyzed reactions
- Ref to **competitive inhibition**:
 - Competitive inhibitor is **structurally similar/similar 3D conformation** to the substrate
 - Competes with substrate for and **binds at / occupies the active site** of the enzyme
 - Effect of inhibition **overcome** by **increasing substrate concentration** → **V_{max} can be reached**
- Ref to **non-competitive inhibition**:
 - **Inhibitor does not have a similar 3D conformation/shape/structure as that of hydrogen peroxide / substrate**
 - **binds to a site away from the active site of the enzyme/ allosteric site**
 - Effect of inhibition **cannot be overcome** by increasing substrate concentration → **V_{max} cannot be reached**

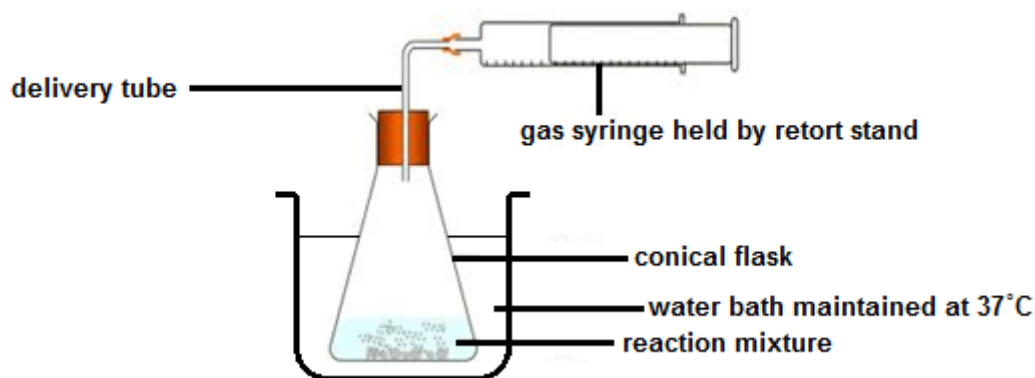
✓ **Experimental Design:** The **rate of reaction** is measured by determining the **average volume of oxygen** gas produced over a **fixed period of time**, **5 min**. **[Method to measure dependent variable]**

3. Variables [2 marks max]:

1. **Independent variable:** **Hydrogen peroxide concentration (%)**; give at least **5 values**; eg **4.0%, 8.0%, 12%, 16%, 20%** **[Range of values with regular intervals and Correct unit]**
2. **Dependent variable:** **Rate of reaction measured by the average volume of oxygen gas produced per unit time ($\text{cm}^3 \text{min}^{-1}$)** **[Correct unit]**
3. Other variables to be **kept constant / Controlled variables (Any 1)**: **[Apparatus & quantity to be indicated]**
 1. **Volume** of potato extract
 - Use a **syringe** to add a **fixed volume** of potato extract solution, **5.0 cm^3** , **from the same stock** (stirred well before use)
 2. **Duration** of reactions
 - A **digital stopwatch** is used to ensure duration of reactions is kept constant at **5 min**
 3. **Temperature**
 - Place conical flask in a **water bath at 37°C** . **Maintain** at this temperature by **adding warm/cool water and use a thermometer to monitor** the temperature of the water bath

4. Volume of 5 mol dm^{-3} Copper (II) Sulphate (CuSO_4) solution (**only for one experiment**)
- **Fixed volume** of 1.0 cm^3 of 5 mol dm^{-3} Copper (II) Sulphate (CuSO_4) solution for the set of experiment conducted in the presence of inhibitor.

4. Labelled Diagram [1 mark]:



5. Procedures [Award only if use of apparatus and quantity stated – Max 4]

1. Using simple dilution, prepare 5 hydrogen peroxide concentrations of 4.0%, 8.0%, 12%, 16%, 20%.

Volume of Distilled water (cm^3)	Volume of 20% hydrogen peroxide stock solution (cm^3)	Concentration of hydrogen peroxide solution (%)
80	20	4.0
60	40	8.0
40	60	12
20	80	16
0	100	20

[1]

2. Set up the apparatus as shown above. **[1 – for 2 equilibration steps]**
3. Using a clean syringe, add 20.0 cm^3 of the **20% hydrogen peroxide solution** into the conical flask.
4. Using a clean syringe, add 1.0 cm^3 of 5 mol dm^{-3} Copper (II) Sulphate (CuSO_4) solution into the conical flask. (Note: volume of inhibitor is very small).
5. Using a clean syringe, add 5.0 cm^3 of potato extract solution into a **test/boiling tube**. (Note: volume of potato extract solution is less than volume of hydrogen peroxide).
6. Leave the conical flask and test/boiling tube in the water bath at 37°C for 5 minutes to allow the temperature to equilibrate. Monitor temperature using thermometer.
7. At the end of 5 minutes, pour the potato extract solution from the test tube into the conical flask. Shake to mix well.

8. Use a delivery tube to connect the opening of conical flask to the gas syringe. Hold the gas syringe using a retort stand. Start the **stopwatch** and allow **1 minute** for the mixture to **equilibrate**.
9. After 1 minute, note the initial volume of air in the 20cm³ gas syringe.
10. Allow the stopwatch to run for **another 5 minutes**.
11. Calculate the volume of oxygen produced after 5 minutes by **subtracting the final volume of air in the gas syringe from the initial volume of air in the gas syringe**
12. **Perform 2 more replicates** by repeating steps 2 to 12, using **hydrogen peroxide and potato extract solution from the same stock** each time.
13. **Repeat** steps 2 to 13 for **hydrogen peroxide concentrations** of **0%, 4.0%, 8.0%, 12%, 16%**.
14. Repeat the entire experiment (steps 2-14) ***without the addition*** of Copper (II) Sulphate. **[1]**
15. **Repeat the entire 2 experiments** (steps 1 to 15) **twice** using hydrogen peroxide and potato extract solution of a different batch. **[1 – for both replicates and repeats]**

6. Control set-ups [1 mark]:

- ✓ **Purpose of control:** The control is set up to show that the **oxygen** production at different hydrogen peroxide concentration is **due to the catalytic action by enzyme catalase and not due to some other factors**.
- ✓ **Control set-ups:** Steps 2 – 12 are performed with and without Copper (II) Sulphate with **all other experimental conditions remained the same** except **replace** the potato extract with **equal volume** of distilled water or boiled and cooled potato extract solution (5.0 cm³).
- ✓ **Expected results:** There should be **no or minimal** production of **oxygen** in the control set-ups.

7. Manipulation of data [1 mark]

- ✓ Calculate the **Average volume of oxygen** produced after 5 minutes for the 3 **replicates**
- ✓ Calculate **Rate of enzyme reaction** as **Average volume of oxygen produced / 5**

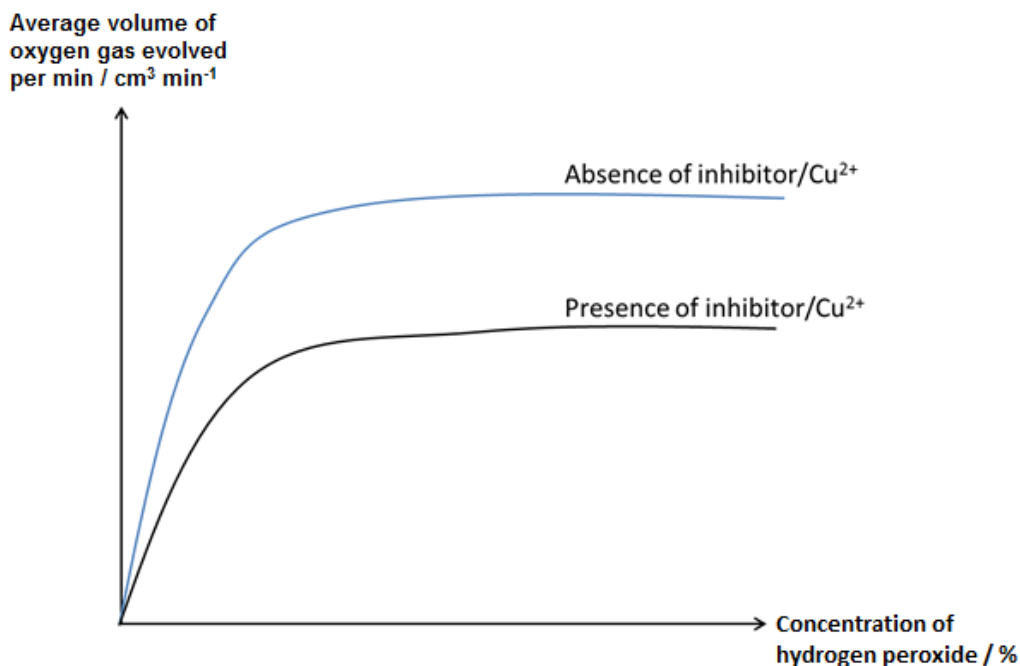
8. Table and Data [1 mark]:✓ **Appropriate headings with units, reject if lines are not drawn**

Concentration of hydrogen peroxide solution (%)	Volume of oxygen gas evolved in 5 min in the <u>presence</u> of Cu^{2+} (cm^3)				Rate of oxygen evolved = Average volume of oxygen gas evolved per min ($\text{cm}^3 \text{min}^{-1}$)
	R1	R2	R3	Average	
20					
16					
12					
8.0					
4.0					
0.0					

Concentration of hydrogen peroxide solution (%)	Volume of oxygen gas evolved in 5 min in the <u>absence</u> of Cu^{2+} (cm^3)				Rate of oxygen evolved = Average volume of oxygen gas evolved per min ($\text{cm}^3 \text{min}^{-1}$)
	R1	R2	R3	Average	
20					
16					
12					
8.0					
4.0					
0.0					

9. Graph [1 mark]:✓ **Labeled axes and correct proposed trend**

Graph of average volume of oxygen evolved per minute ($\text{cm}^3 \text{min}^{-1}$) against concentration of hydrogen peroxide (%).



10. Risks & Precautions [1 mark]:

Risk	Precaution
Hydrogen peroxide is corrosive / Copper sulfate is an irritant to the skin	Wear safety goggles to prevent any possible eye contact with hydrogen peroxide/ Use protective gloves to prevent direct skin contact with hydrogen peroxide/ Immediately flush the eye / wash with water when the chemical reagent comes into contact with the eyes and/or skin.
Mercury is toxic and can lead to poisoning .	Exercise caution in handling thermometer so as to prevent breakage.

Question 5

- (a) With reference to genetic diseases, describe cystic fibrosis and its treatment using gene therapy. [10]

Cystic fibrosis as a genetic disease [max 6 marks]

- Autosomal recessive** disorder;
- a mutation of Cystic Fibrosis Transmembrane Regulator (CFTR) gene on chromosome 7;**
- a deletion of 3 nucleotides/bases of (CFTR) gene,** resulting in the **loss of phenylalanine** in polypeptide;
- change in 3D conformation of CFTR protein,** resulting in a **non-functional CFTR protein**/ channels no longer able to **transport chloride ions**;
- normal efflux of chloride ions out of lung epithelial cells is prevented,** chloride ions **accumulate in the epithelial cells,** attracting water into the epithelial cells;
- causing **thick mucus** that clogs **airways/ lung respiratory tracts**;
- leading to breathing difficulties/ resulting in bacterial lung infections/ pneumonia/ permanent lung damage/ formation of scar tissue (fibrosis) and cysts in the lungs;**
- thick mucus** also blocks the **duct of the pancreas/** reduces secretion of pancreatic enzymes, leading to **digestive problems/ malnutrition.**

Treatment using gene therapy [max 6 marks]

1. **[Viral vector method]:** The **normal CFTR gene** is inserted into **modified/ inactivated adenovirus** vector where it is **no longer virulent** so that it does not replicate in host cells;
2. The **recombinant adenovirus** vector carrying **normal CFTR gene** is introduced into target cells which are the **lung epithelial cells** in the patient by **direct injection**;
3. **Viral glycoproteins** bind to specific **receptors** on surface membrane of target lung epithelial cells;
4. Virus enters by **endocytosis**: surface membrane of target cell **invaginates** and pinches off to form intracellular **vesicle (endosome)**;
5. **Viral envelope fuses with endosomal membrane**, nucleocapsid is released into the cytoplasm;
6. the **viral DNA with normal CFTR gene enters nucleus** through the pores in the nuclear envelope;
7. The **normal CFTR gene is expressed**, producing **functional CFTR protein** that is embedded within the cell membrane.

OR (DESCRIBE ONLY 1 method)

8. The **normal gene** is inserted into **liposomes**;
9. The liposomes are **directly injected into the patient's body** or **inhaled in an aerosol** using **nasal sprays**, so as to deliver the corrective CFTR allele directly to **lung epithelial cells**;
10. **phospholipid bilayer** of liposome mimic **structure of cell surface membranes**;
11. Liposome **fuses** with the **cell surface membrane** and delivers the **normal allele** into the cytoplasm.
12. The **normal CFTR gene is expressed**, producing **functional CFTR protein** that is embedded within the cell membranes.

(b) Discuss the ethical and social considerations of gene therapy.

[5]

Issues on the safety and efficacy of gene therapy

1. Presence of **limitations in techniques/uncertainties of outcomes** as well as **potential clinical risks**, which may lead to **unknown long term consequences** to the well-being of the recipients.
2. It is difficult to determine the **long term effects of exposure to viral vectors** on the human genome.

Extremely high cost

3. Gene therapy is currently very expensive, require expertise and equipment found only in major medical centres, concerns about gene therapy become a **luxury available only to the rich and powerful**.
4. This could possibly lead to **social disparity between the rich and the poor**, with the rich having better accessibility to treatments.

Potential impact on improving patients' quality of life

5. Gene therapy might **provide alternative forms of treatments** directed at currently unmet clinical needs, where conventional treatments have failed.
6. **Germline therapy** can eradicate disease from subsequent generation and prevent transmission of serious genetic diseases from parent to offspring

Potential practice of eugenics

7. **(non-therapeutic enhancement possibilities)**, for example, improving memory and intelligence by genetic manipulation
8. concerns about the widespread use of gene therapy making **society less accepting of people** who have mild disorders / genetic diseases or less able;
9. raises ethical issues such **type of circumstances that determines whether genomes should be altered** and the **decision of what is a disability or disorder**.

Interfere with evolution

10. elimination of unwanted alleles from the gene pool will decrease **genetic variation**;
11. **detrimental to the survival of our species in future**
/ genes that are damaging under some conditions may be advantageous under other conditions / germ-line gene therapy would forever change the genetic makeup of an individual's descendants / the human gene pool would be permanently affected

Interfere with life

12. Altering ones' genes and changing the way one was created go against many **religious and cultural beliefs**.
13. The **unborn child** who would be affected by germline gene therapy **cannot choose** whether to have the treatments

- (c) Using named examples, discuss how genetically modified organisms may solve the demand for food in the world. [6]

1. **Bt Corn: genetically modified corn with the insertion of a gene from *Bacillus thuringiensis* into the corn.**
 2. This gene codes for **Bt delta endotoxin**.
 3. **The endotoxin, kills a common pest that destroys corn crops, called the European Corn Borer (a caterpillar).**
 4. **Farmers growing Bt corn do not need to spray insecticides on their crops to get rid of the pests.**
 5. **Since the crops are not infested with pests, this allows the farmers to increase their yields, productivity, improve their livelihoods and alleviate poverty.**
-
1. **Genetically Modified Salmon/ The Atlantic salmon has been genetically modified with two DNA sequences.**

2. They are:
 - A promoter of an anti-freeze gene from an ocean pout.
 - A growth hormone gene from Chinook salmon.
3. These 2 DNA sequences enable the salmon to continue producing growth hormone all year round, even during the winter months.
4. Hence, GM salmon can grow 5 times faster than its wild type counterpart/ faster growth rate
5. Because of the shorter maturation time, GM salmon raised in tanks can be harvested and sold in a shorter time and thus increasing the profit.